Speaker verification based on fusion of acoustic and articulatory information

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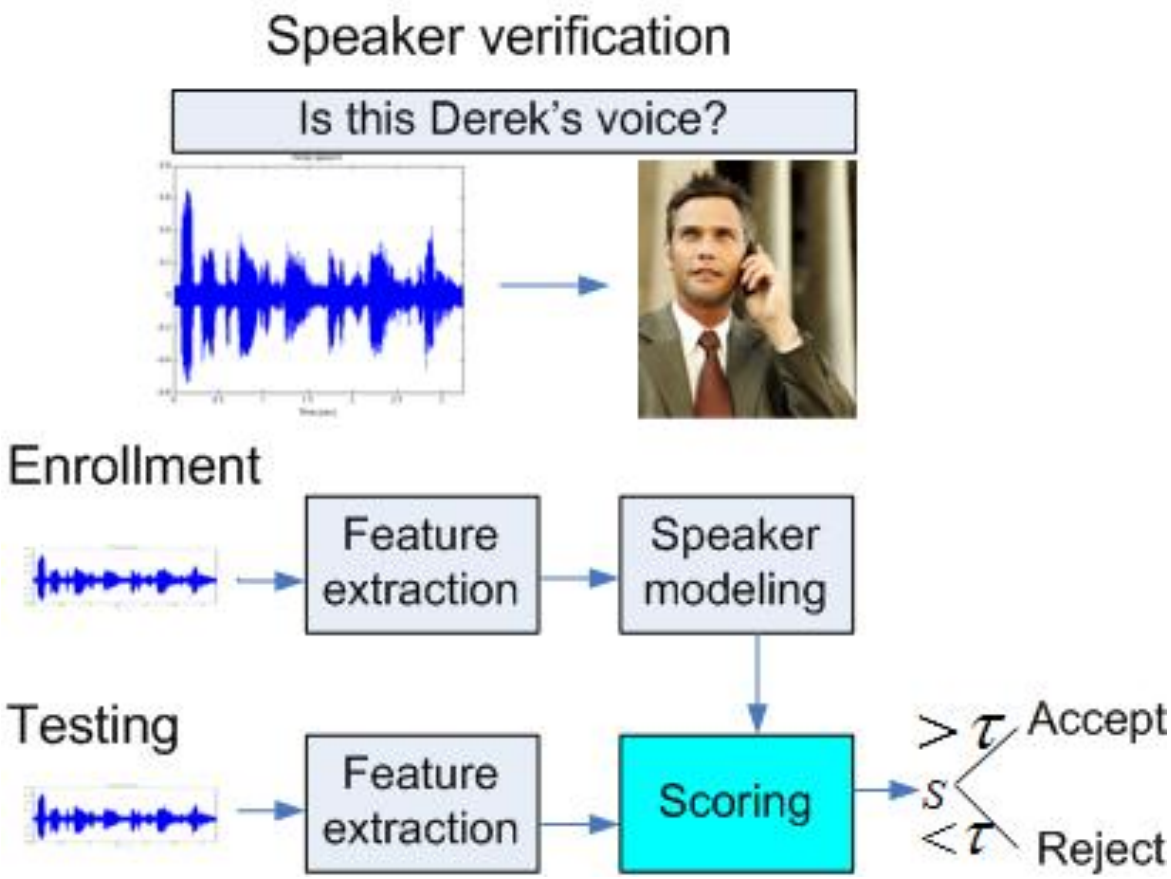
Speaker Verification

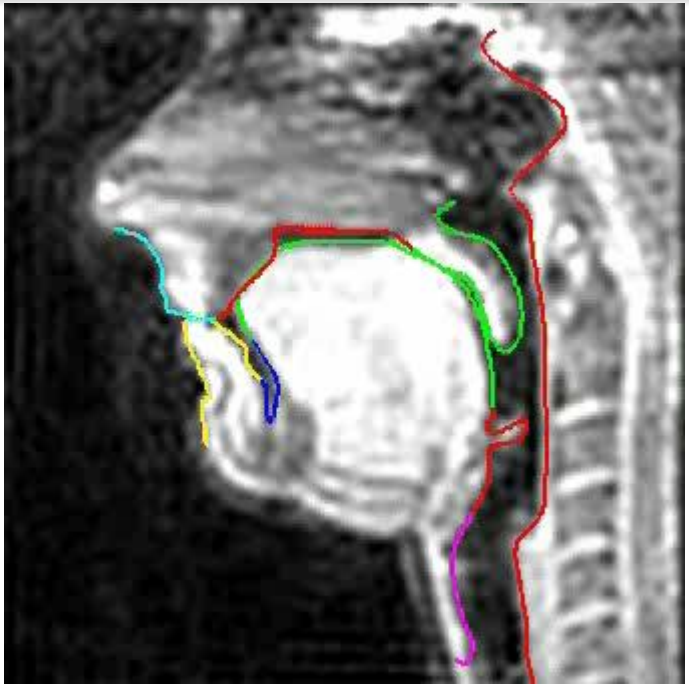
- Speaker verification

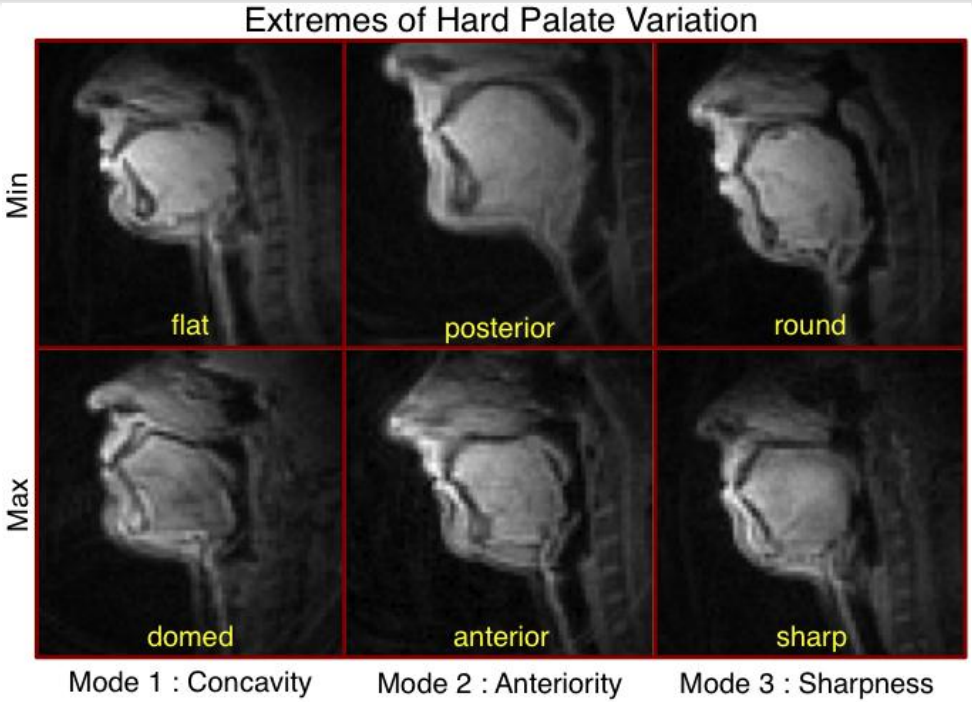


Speech production for Speaker ID

- Speaker verification
  - Acoustic level methods
    - Joint factor analysis (JFA) (Kenny, 2007)
    - I-vector (Dehak, 2011)
    - Simplified Supervised I-vector (Li, 2013)
    - Within class covariance normalization (WCCN) (Hatch, 2006)
    - Probabilistic linear discriminative analysis (PLDA) (Prince, 2007, Matejka, 2011)
  - Feature level or score level fusion based on multiple features
    - Short-term spectral features (MFCC, LPCC, PLP, etc.)
    - Spectral-temporal features (FDLP, Gabor, etc.)
    - Prosodic features (pitch, energy, duration, rhythm, etc.)
    - Voice source features (glottal features)
    - High level features (phoneme, semantics, accent, etc.)
  - Apply features from the speech production system?

Morphological variability





- Vocal track morphological variability
  - Vocal tract length (Peterson, 1952; Fant, 1960, Lee, 1999, Stevens, 1998)
  - Shapes of hard palate and posterior pharyngeal wall (Lammert, 2011)
    - "Automatic Classification of Palatal and Pharyngeal Wall Shape Categories from Speech Acoustics and Inverted Articulatory Signals", Li et.al, Interspeech satellite workshop SPASR, 2013.
  - We believe that in order to make standard pronunciation, articulation need to compensate the vocal tract morphology.

Articulatory variability

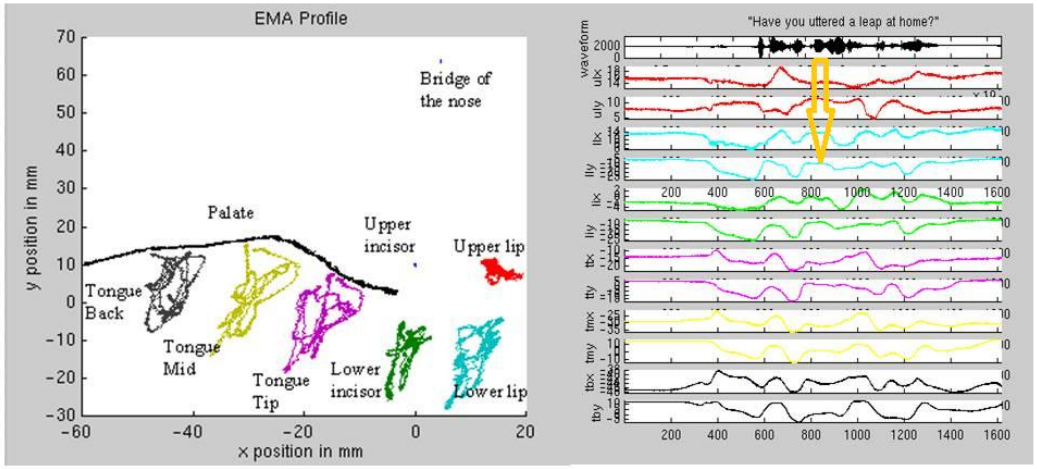
- Articulation also contains speaker specific information
  - Flat palates exhibit less articulatory variability than highly domed palates during vowel production (Perkell, 1997; Mooshammer, 2004; Brunner, 2005; Brunner, 2009)
  - Articulation of coronal fricatives is influenced by palate shape
    - apical vs. laminal articulation of sibilants (Dart, 1991)
    - jaw height and tongue body positioning (Honda, 2002, Thibeault, 2011).
  - An example in Singing

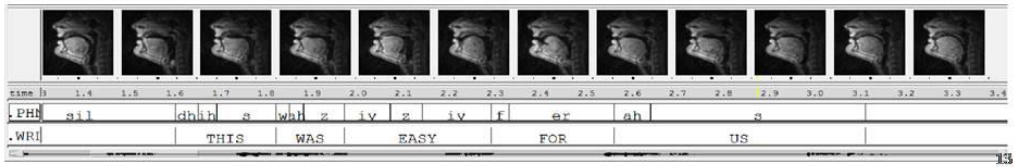
- Different speakers articulate even the same words differently

Acoustic-to-articulatory inversion

- For Speaker ID
  - Real articulation measurement impossible
  - Speaker independent acoustic-to-articulatory inversion (Ghosh, 2010)
  - Map inter-speaker acoustic variability to the reference speaker's intra-speaker articulatory variability

RAW MEASUREMENT FEATURES

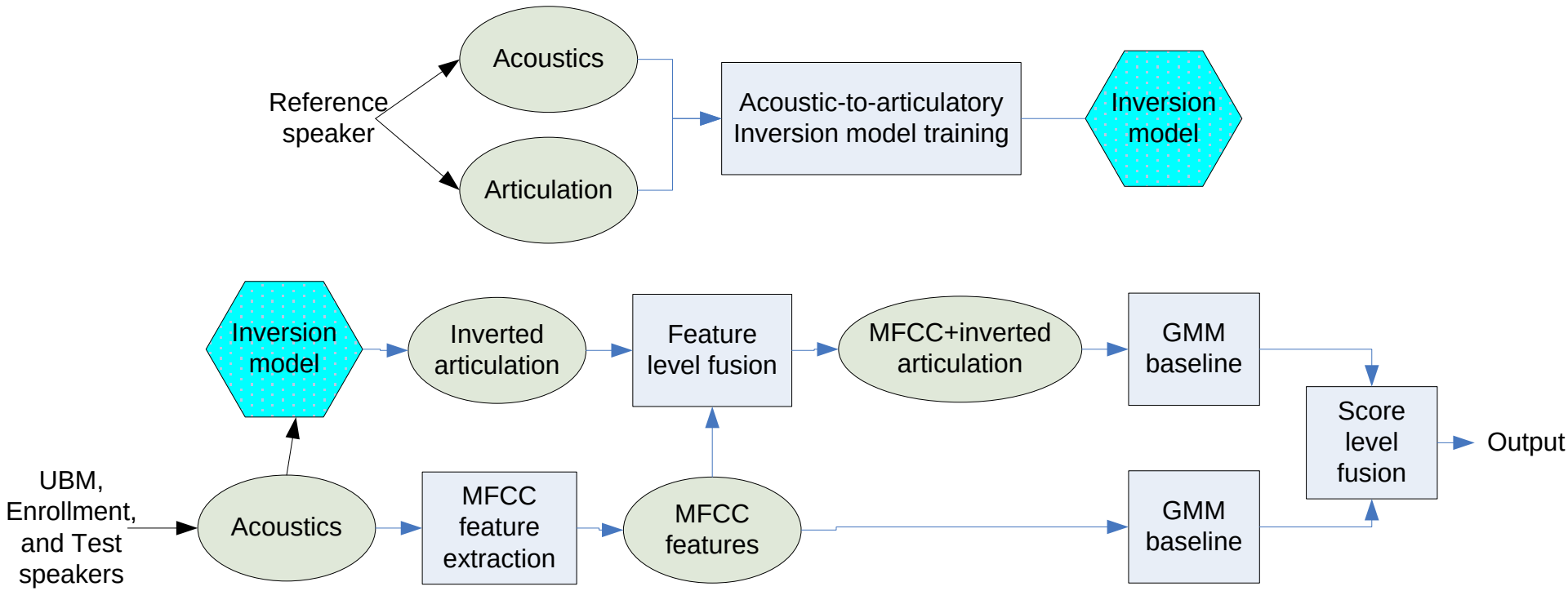




P. K. Ghosh and S. S. Narayanan, "A generalized smoothness criterion for acoustic-to-articulatory inversion," JASA, 2010.

System Overview

- Feature level fusion and score level fusion



Front end processing and GMM baseline

- Front end processing
  - Wiener filter applied on the XRMB data
  - Real or inverted articulatory data sampled at 100hz
  - 25ms window size with 10ms shifts for MFCC extraction
  - 36 dim MFCC (18dim + delta) with MVN
  - MVN on the real articulatory data not on the inverted one
    - Real articulation (mean/var) has encoded vocal tract shape information
    - Remove mean/var of the real articulation for fair comparison
    - Inverted articulation (mean/var) has rich speaker information
  - Concatenating MFCC and articulation together as feature level fusion
- GMM baseline
  - Conventional GMM-UBM-MAP approach (limited data)
  - GMM size 256, relevant factor 16, AT-norm

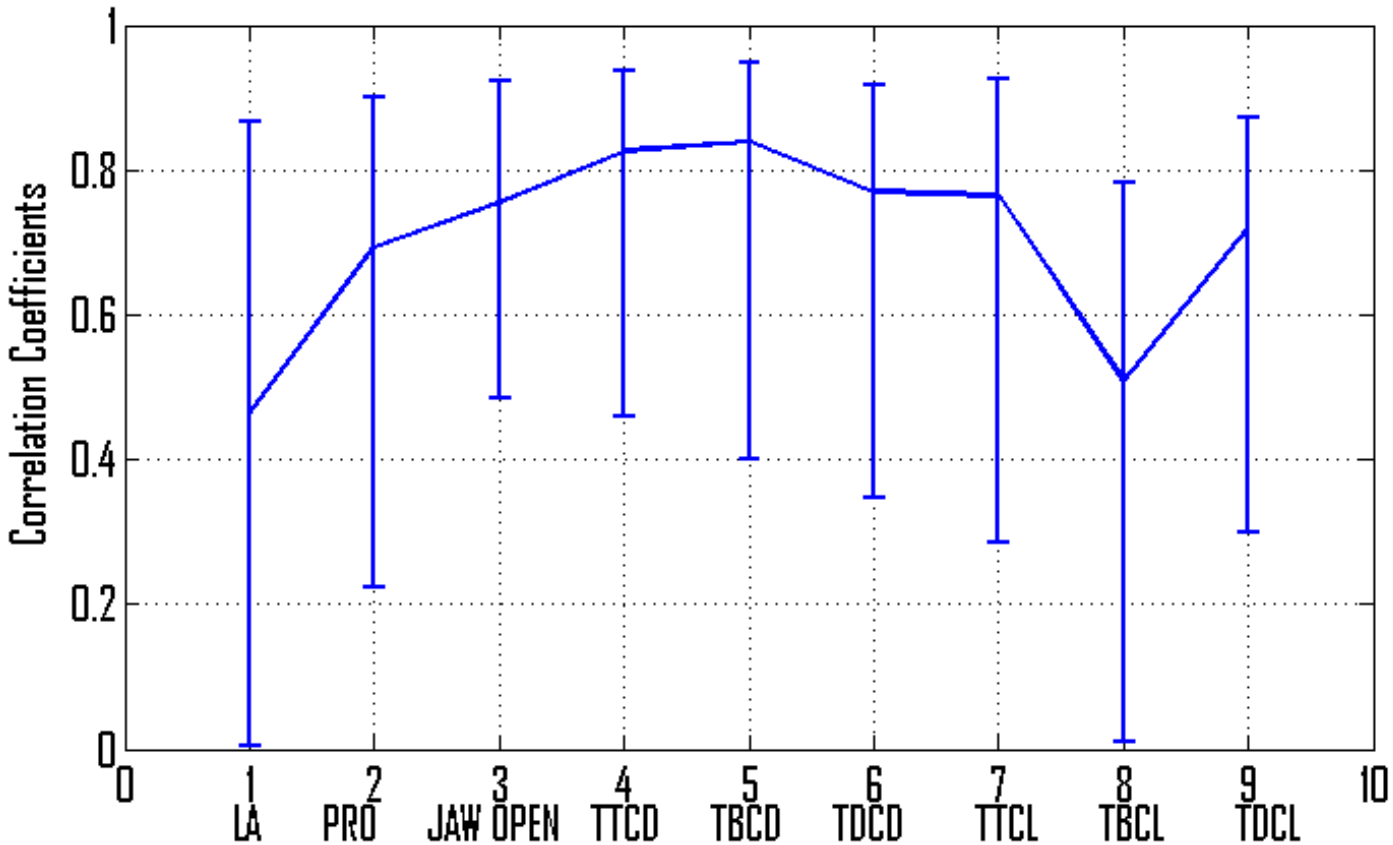
Database and experiment design

- Wisconsin X-Ray Microbeam data (XRMB) (Westbury,1990)
  - Both articulatory measurement and simultaneously recorded speech signal are available from multiple speakers
  - Clean speech (okay for inversion, MRI data too noisy)
  - Session 1-101, speaker JW11-63, 46 speakers, 4034 utterances
  - Average duration of 5.72 seconds per utterance.
  - Two protocols: ALL and L5S (longer than 5s data for testing)

Data sets & Protocol	ALL	L5S
Background: all sessions of JW11-40	√	√
Target: session 11 of JW41-63	√	√
Test: other sessions of JW41-63	√	
Test: other L5S sessions of JW41-63		√
Tnorm: sessions 11,12,79,80,81 of JW11-40	√	√

Experimental Results (1)

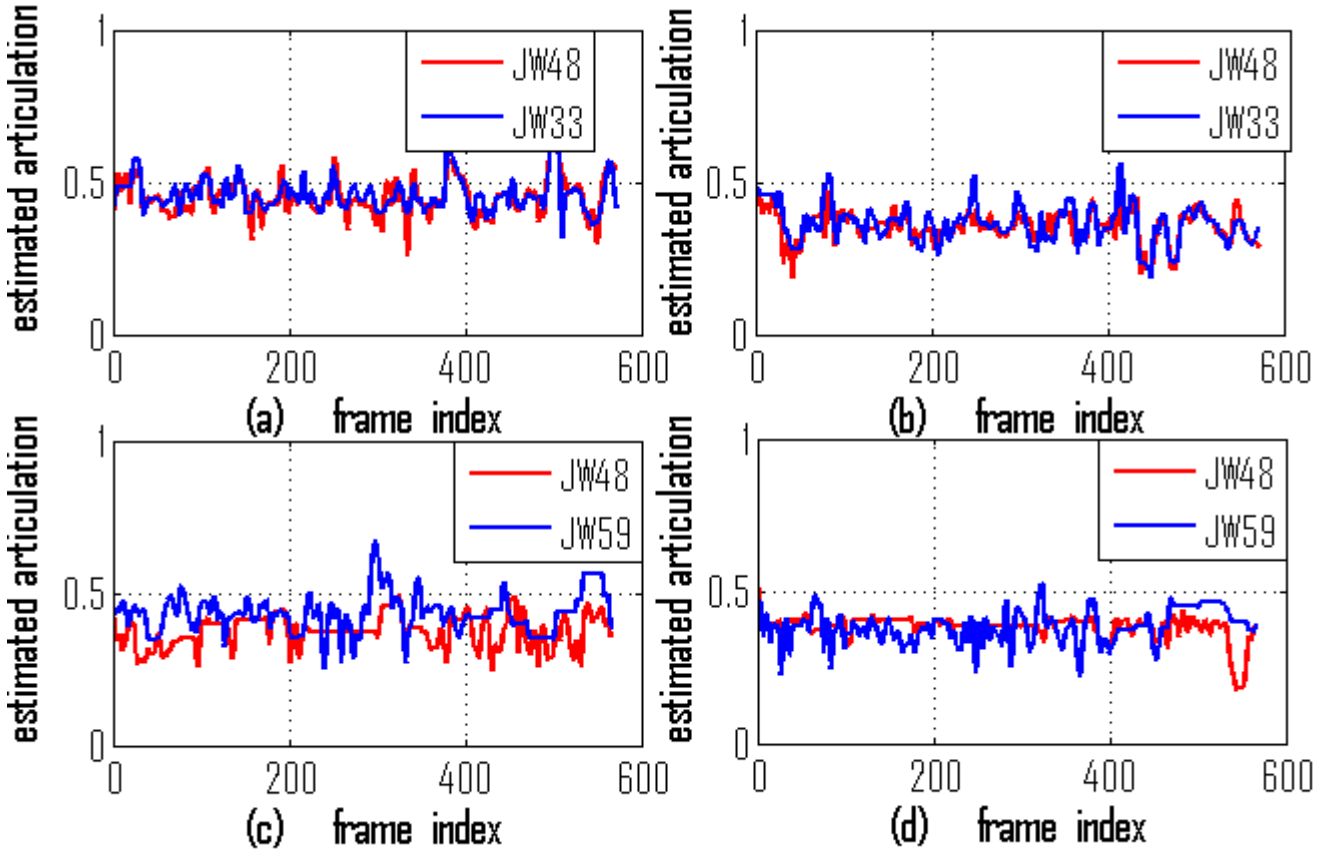
- Error bar of pair-wise correlation coefficients between session one estimated articulatory signals (after DTW) from all 46 speakers



- Lip aperture (LA), lip protrusion (PRO), jaw opening (JAW OPEN), the constriction degree (CD) and constriction location (CL) of tongue tip (TT), tongue blade (TB), and tongue dorsum (TD).

Experimental Results (2)

- Estimated articulatory signals of lip aperture (LA) and tongue body constriction location (TBCL) from two-speaker pairs (session 1).



- JW48 and JW33 are both female and with the highest correlation in the previous figure
- JW48 and JW59 are from different genders, mean and variance have speaker information

Experimental Results (3)

- The performance of 26 speakers (closed set) identification systems based on different utterance-level features derived from estimated articulatory data
  - Multi-class SVM
  - Train: sessions 12, 79, 80 and 81 of all 26 speakers in the background data set
  - Test: session 11 of all 26 speakers in the background data set

Features & Systems	1	2	3
mean	√	√	√
variance		√	√
mean crossing rate			√
Accuracy	32%	48%	52%

Experimental Results (4)

- Performance of MFCC-real-articulation system, “all-small” protocol
  - “all-small” protocol : same as “all”, but a subset of real articulatory data were removed from the data sets due to the missing data issue in some channels
  - Feature level fusion with real articulation data helps (mean/var normalized)
  - Score level fusion achieved big EER reduction

ID	Systems	“All-small” protocol	
		OptDCF	EER
1	MFCC-only	11.04%	11.95%
2	MFCC-real-articulation	9.98%	10.15%
3	**Score level fusion 1+2**	**6.42%**	**6.77%**

Experimental Results (5)

- Performance of MFCC-estimated-articulation system, “all” protocol

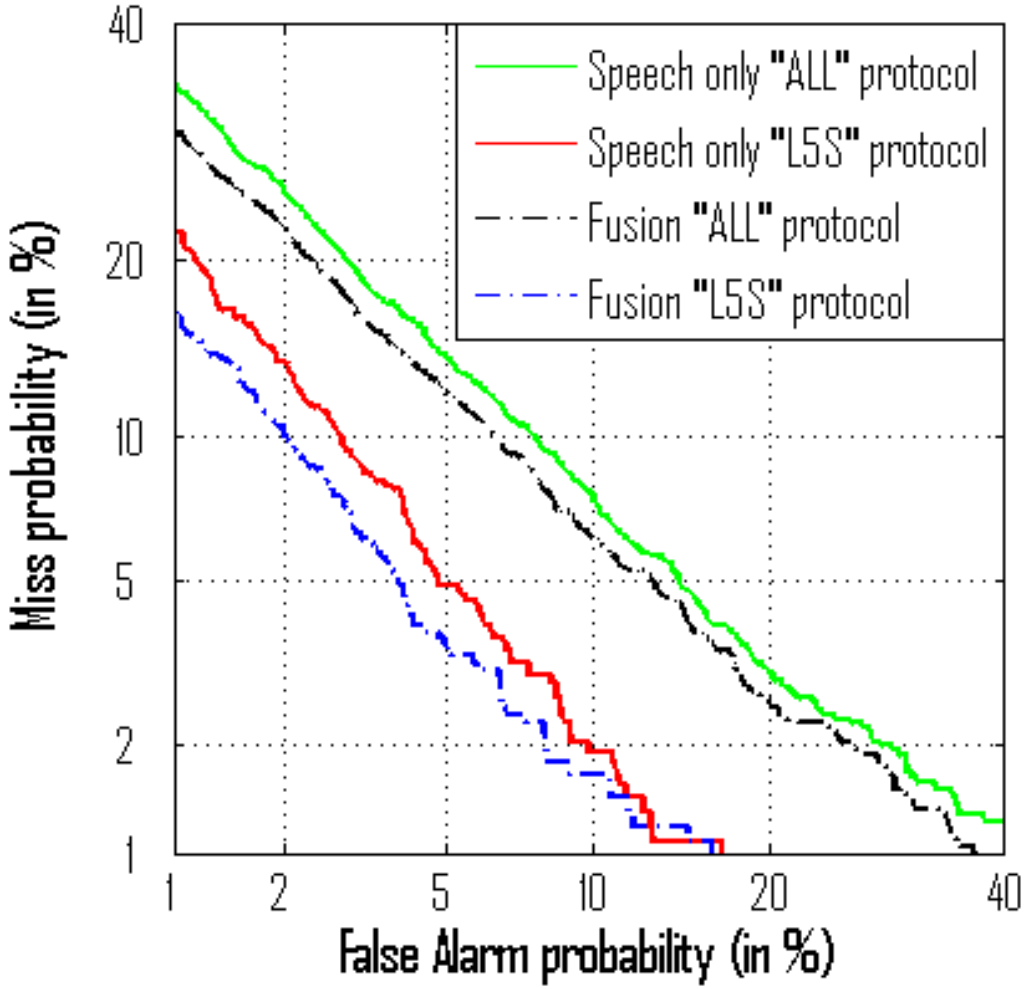
ID	Systems	“All” protocol		
		OptDCF	EER	Accuracy
1	MFCC-only	8.68%	8.73%	89.65%
2	MFCC-estimated-articulation	8.40%	8.44%	90.92%
3	**Score level fusion 1+2**	**7.83%**	**7.91%**	**91.74%**

- Performance of MFCC-estimated-articulation system, “L5S” protocol

ID	Systems	“longer than 5s” protocol		
		OptDCF	EER	Accuracy
1	MFCC-only	4.84%	4.88%	95.95%
2	MFCC-estimated-articulation	9.34%	4.52%	97.14%
3	**Score level fusion 1+2**	**4.05%**	**4.17%**	**97.02%**

Experimental Results (6)

- DET curves of the MFCC only system and the score level fusion system for “ALL” and “L5S” protocols.



Conclusions

- We propose a practical fusion approach for speaker verification using both acoustic and articulatory information
- Significant performance enhancement (40% relatively) by concatenating articulation features from measured articulatory movement data with MFCC
- Moderate gains (9%-14% relatively) using estimated articulatory features obtained through acoustic-to-articulatory inversion
- Future works cover investigating better inversion methods and evaluating the proposed methods on NIST SRE database.